Main node TH PCB (hw/main, rev 2.0): 30-pin ESP32 DevKit = relays + car inputs + GPS + IMU + CAN, under-dash (D46).

Main node



File: main-node.kicad_sch

Power: power module #2 (F1 2 A) fed from RJ45 pins 4/5 + 7/8 via J39; 3V3 into the DevKit 3V3 pin (J30).
3V3_SW (Q30, GPIO32 low = on) feeds CAN transceiver + GPS + MPU6050 (~60 mA), off in sleep.
R64 120 ohm = bus end 2 termination. U30/C35/D39 (SMD) placed by JLCPCB. GPS UART2 (GPIO16/17), MPU6050 I2C (21/22).
Inputs IG1/HEAD/REV (J31) via inverting NPN stages; HEAD has a 100k pull-up from COIL+.
Relay drivers: 4 x BC337-40 low-side + 1N4007 flyback to COIL+ (J32 in from module J3, J34 out to the relay box).

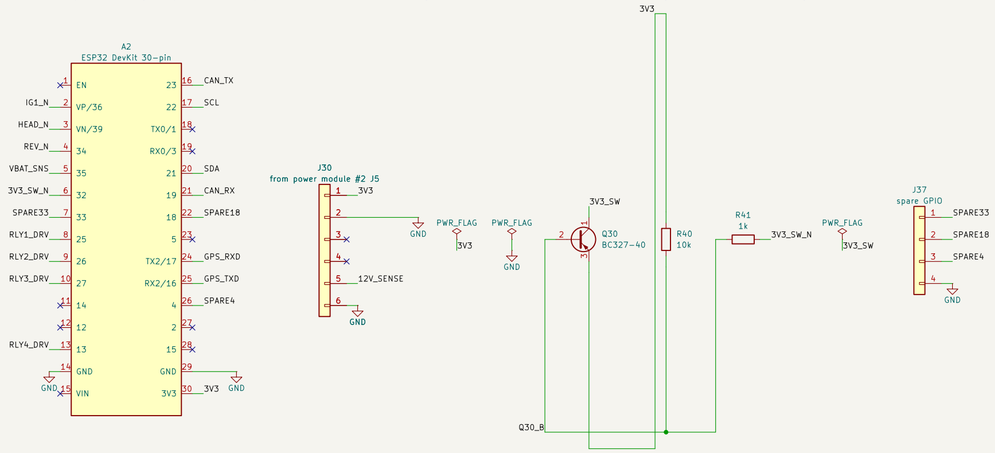
Embedded symbols (lib PERF); generated by scripts/gen-perfboards
Source: docs/design-main.md (2026-10-29, arch rev 4.3)
Miata NA hardware — TH PCB (owner-fitted TH, JLC SMD)

Sheet: /
File: main.kicad_sch

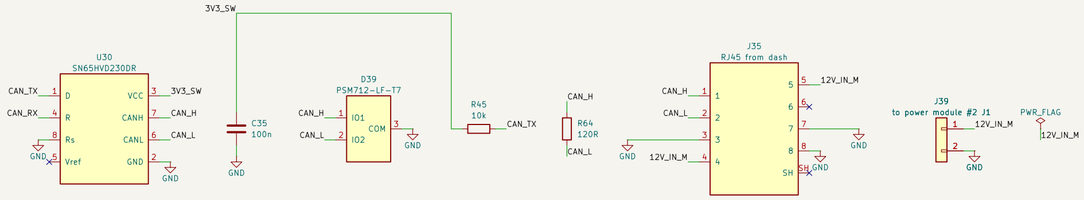
Title: Main node TH PCB (root)

Size: A3	Date: 2026-10-29	**Rev: 2.0**
KiCad E.D.A. 10.0.7		Id: 1/2

ESP32 30-pin DevKit (3V3 pin fed from power module #2 J5 via J30), switched 3V3 for CAN + GPS + IMU (Q30, GPIO32 low = on)



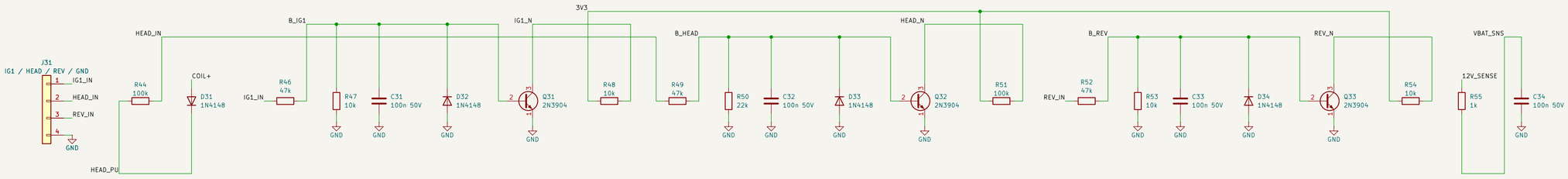
CAN: SN65HVD230DR + 100n + PSM712 (JLC), R64 120 ohm (bus end 2); RJ45 from the dash: 1/2 CAN, 3 GND, 4/5 +12 V, 7/8 GND -> J39 -> module #2 J1



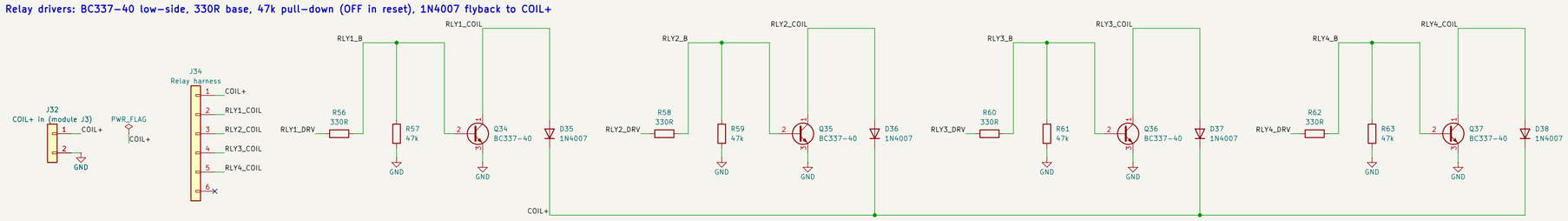
GPS GY-GPS6MV2 on UART2 (1k series both ways), IMU GY-521 on I2C (module pull-ups, AD0 = GND -> 0x68)



Car inputs (J31): IG1 -> GPIO36, HEAD -> GPIO39 (100k + 1N4148 pull-up from COIL+, 22k/100k: ~0.1 mA at rest), REV -> GPIO34; VBAT GPIO35



Relay drivers: BC337-40 low-side, 330R base, 47k pull-down (OFF in reset), 1N4007 flyback to COIL+



Key values (docs/design-main.md)

Q30: I_B 2.27 mA, load 60 mA -> forced beta 26 << 250, 3V3_SW - 3.19 V
Relay driver: I_B = (2.64 - 0.85)/330 = 5.4 mA, coil 16 V / 80 ohm = 0.20 A -> forced beta 37 < h_FE(300 mA) 60
HEAD stalk open: V_B = 11.3 x 22/169 = 1.47 V -> on, 0.067 mA + 0.033 mA collector = 0.1 mA at rest; stalk to GND -> off
Inputs: threshold 3.7 V, f_c 193 Hz (design-dash 6)
CAN: 120 ohm (R64) || 121 ohm (ME221) = 60.2 ohm; R64 dissipation (2.5 V)^2/120 = 52 mW dominant

Embedded symbols (lib PERF): generated by scripts/gen-perfboards
Source: docs/design-main.md (2026-10-29, arch rev 4.3)
Miata NA hardware — TH PCB (owner-fitted TH, JLC SMD)

Sheet: /Main node/
File: main-node.kicad_sch

Title: Main node TH PCB — ESP32, CAN, GPS, IMU, inputs, relay ...

Size: A2	Date: 2026-10-29	**Rev: 2.0**
KiCad E.D.A. 10.0.7		Id: 2/2